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DIGITAL PRODUCT PASSPORT · MULTI-BODY STEWARDSHIP

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ISO/IEC JTC 5 · DIGITAL PRODUCT PASSPORT

An international framework for the Digital Product Passport

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UniDPP — convened under CalConnect, multi-body stewardship

01 · INTRODUCTION

Scope and submission context

ISO/IEC JTC 5, *Digital Product Passport*, inherits two partially incompatible asset bases: the European EN 18216–18246 series prepared by CEN-CLC/JTC 24 under the Ecological Design for Sustainable Products Regulation (ESPR), and AWI ISO 25534-1, which splits into a vocabulary Part 1 and a Part 2 addressing the passport itself. A third body of practice — the IDTA submodel templates, the UN Transparency Framework (UNTP), and national platforms such as the GS1 China commodity QR-code service — is already deployed in the field.

This paper states the joint position of the drafting consortium in one sentence: **JTC 5 should standardize the neutral mechanism by which many legal regimes share one product record, and should decline to make any single regime the universal envelope.** The position is developed from a complete converted corpus of the eight ENs, two CWAs, 77 IDTA specifications, AWI 25534-1 and the JTC 5 plenary document set, and from running reference-implementation work (UniDPP) reported separately as paper 4 of this series.

Four companion contributions accompany this paper: semantic registry requirements (paper 2), lifecycle events and validity (paper 3), the reference implementation report (paper 4), cross-recognition trust (paper 5) and the terminology dataset announcement (paper 6).

02 · THE PROBLEM

Four findings that define the work

2.1 The core noun is normatively contested

The EN series defines the DPP as a *"digital record of product characteristics throughout its life cycle"*. AWI 25534-1 defines it as *"a concept or technology which enables sharing of product related information"*. One definition names a record; the other names a capability. Both texts are normative in their respective systems, and neither cites the other. Terminology is therefore not a housekeeping item: the two systems cannot interoperate until the object of standardization is agreed. The consortium addresses this in paper 6 with a harvested terminology dataset proposed as the single source of truth for a Part 1 (Vocabulary).

2.2 The semantic layer is a named vacancy

EN 18223 §4.3 requires a governed, "regulation-based" semantic repository, and its examples are placeholder URLs (`dictionary1.eu`). The operator, the resolution protocol, caching, offline behaviour and failure semantics are unspecified. Every `dictionaryReference` in the series points into this vacancy. Paper 2 proposes the filling: ISO 19135:2025 federated register governance, the UNTDED data-element seed, a resolution protocol, and CDDAL canonicalization.

2.3 The event layer is mandated but not modelled

EN 18223 §4.2 mandates change logging (identifier, timestamp, actor, changed properties) and §4.1.2.1 gives example values for `dppStatus` (active / inactive / archived / invalid) — but the series defines no event types, no state machine, no transition rules, and no validity semantics. The legally decisive field of the passport is undefined. Paper 3 supplies the 16-event post-sale taxonomy and the state machine.

2.4 One product, many jurisdictions — by design, not by exception

A product placed on the world market is simultaneously subject to EU delegated acts (textiles 2026; the registry implementing act due 30/06/2027; steel 30/07/2027), China's CCC and China-RoHS regimes under GB standards, Japan's METI/PSE/Top Runner rules, Korea's KC certification, United States state and federal-procurement rules, and UKCA. A single physical passport must therefore carry **many jurisdiction profiles at once, each independently governed, trusted, versioned and served**, without any region being the universal envelope. This is not a new problem: trade documentation has solved it repeatedly, and the precedents are citable:

- **ePassport (ICAO 9303)** — one document, per-state inspection, and a CSCA **master list** of trust anchors for cross-recognition;

- **ISO 20022** — a global repository with per-jurisdiction validation rule profiles;
- **DCAT-AP** — a core specification with governed national profiles;
- **UNTDED** — the United Nations Trade Data Element Directory, whose historic regional subsets are ISO/TC 154's own decades-long exercise in national profiling of a neutral base;
- **ISO 10303 (STEP)** — core plus composable application modules;
- **GS1 Digital Link resolvers (RFC 9264)** — linksets with per-context routing; and **ETSI/eIDAS trusted lists** — jurisdictional trust infrastructure with defined cross-use.

The EN series itself concedes the principle intra-EU: EN 18223 places the data of all applicable regulations in one passport. The international framework globalizes that principle symmetrically: every jurisdiction's regulation becomes a governed profile of one neutral core.

03 · DESIGN INVARIANTS

Fourteen non-negotiables

The framework is specified as invariants — properties every conforming implementation must hold — rather than as implementation choices. The fourteen invariants are grouped in five families.

ID	Family	Invariant
I1	Identity & semantics	One subject, one identity, never re-minted; identifiers may be dormant before use and are later adopted, not replaced.
I2	Identity & semantics	Single definition, many constraints: profiles constrain and transform, never redefine; inter-profile disagreement is expected and auditable; same-class contradiction is an integrity incident.
I3	Identity & semantics	Inputs are recorded at the finest identifier granularity available; dormant identifiers are permitted always.
I4	Structure & lifecycle	Append-only event sourcing: the distributed, log-anchored event log is the authoritative record; no single party holds it alone.
I5	Structure & lifecycle	Typed relationship algebra: association, derivation, installation, type-lineage, custody, membership and same-subject correlation edges carry distinct legal semantics — one physical thing may carry several sovereign passports, correlated never consolidated (clause 6.3 k).
I6	Structure & lifecycle	State machine legitimacy: normative statuses, defined transitions, dated applicability including retroactive effect; profile growth never changes identity.
I7	Structure & lifecycle	Passports are issued where legal duty attaches; issuance, hosting, identity, anchoring, archiving and trust are separable duties, and republication never transfers authority.
I8	Projection & trust	The passport is a deterministic, registered, legally-cited projection of the testimonial record; transforms publish uncertainty, rounding and decision rules; no pre-transformed duplicates are stored.
I9	Projection & trust	Graded trust: trust markers, multi-suite signatures, verdicts with coverage reports, three verification readings, revocation reason determines retroactivity, taint propagates through the graph.
I10	Sovereignty & neutrality	Profiles are jurisdiction × sector × characteristic, predicate-triggered, each binding data points, transforms, cryptographic suites, trust, access, presentation and carriers.

I11	Sovereignty & neutrality	No load-bearing proprietary layer: mount, identity, semantics and trust are open ISO/IEC/UN instruments; any proprietary metamodel is confined to informative mappings.
I12	Sovereignty & neutrality	Enumeration resistance: the system is verifiable without being browsable; blind edges by default; proof-of-binding does not imply knowledge-of-parent; logs anchor commitments, not facts.
I13	Operation & resilience	Degradation ladder: Tier A offline minimum, Tier B served, Tier C archival; freshness verdicts degrade explicitly; availability is decoupled from origin servers.
I14	Operation & resilience	Continuity: successions exist for hosts, authorities, schemes, algorithms, profiles and regimes; identities outlive their issuers and hosts.

The invariants are deliberately implementation-neutral: they constrain what any conforming system must preserve, not how it is built. The corresponding failure evidence — every defect class observed across the corpus — reduces without exception to a violation of one or more of these invariants at a layer seam.

Two invariants deserve plain statement. **I12, enumeration resistance:** the system is **verifiable without being browsable** — resolution by identity on request, traversal by per-edge rights, blind edges by default, and logs that anchor commitments (hashes), never facts; nobody can enumerate the installed base of anything, and recalls run by locally evaluated published predicate, never traversal-push. **I13, the degradation ladder:** every capability degrades cleanly to the document-lookup minimum (carrier, resolver, served page) through Tier A (offline minimum), Tier B (served) and Tier C (archival), with staleness marked explicitly, never silently.

Where the layers join, the system cracks; the framework therefore fixes twelve **seams**, and every defect in the corpus defect register (AUDIT) is a seam failure: S1 carrier↔resolver (code-format translation; GM2D transition); S2 resolver↔hosts; S3 core↔profiles — the mount, where neutrality is enforced; S4 profile↔registry; S5 registry↔registry (FERIN federation, CDDAL canonicalization); S6 event-log↔transparency-logs (commitment anchoring, salt discipline); S7 issuer-credentials↔profile-trust-lists; S8 transforms↔engines; S9 parent↔child passports; S10 device↔system; S11 Tier A↔Tier B; S12 passport↔adjacent regimes (customs/WCO, e-invoicing, EPCIS/UNTP, CIDOC-CRM, ISO 20022).

04 · THE LENS AND PROFILE MODEL

How many jurisdictions share one record

The framework's central abstraction is the **profile as a calibrated lens placed on the product's testimonial record** (the digital twin). The scene — the append-only record of what is known about the subject — is never altered by observation. The lens — a registered, versioned, signed object — selects and transforms it: focal length is granularity (model to instance zoom); depth of field is traversal depth; the colour filter is the classification transform (the same product is energy class "A" in one jurisdiction and "B" in another, and the difference is published as decision rules, never hidden); filter stacking is profile composition; the published uncertainty budget is a mandatory lens property; and the **mount** — the profile interface on the neutral core — is the thing JTC 5 would standardize, so that any registered lens by any maker clicks on. A profile is thus not passive glass but a **legal measuring instrument**: pattern-approved at registration, calibrated by version and cited rules, periodically re-verified by conformance runs — the governance vocabulary of legal metrology, applied to data projections.

Concretely, a profile is a registered, versioned object binding:

- profile identifier and owning regulator, with legal basis and effective dates;
- a data-point list referencing registry items (never inline definitions);
- validation rules and executable transform bindings (unit conversion, derivation, aggregation, classification, localization mapping) with published uncertainty, rounding and conformity decision rules;
- languages, presentation and carrier requirements;
- trust requirements (accepted suites, trust lists) and access policies in terms of the EN 18239 role model;
- a **trigger predicate** — profiles attach by predicate on product facts (jurisdiction of sale, sector, and characteristics such as species content, heritage status or age thresholds), including time predicates that fire dated applicability.

Composition is three-axial: **jurisdiction × sector × characteristic**. The third axis deserves statement. **Characteristic profiles** attach by predicate on product facts — age, material content, heritage status, market status — rather than by law-of-place or industry, splitting into **regulatory-triggered** profiles (CITES species content, cultural-goods import control, conflict-minerals origin) and **voluntary value profiles** (provenance and grading for antiques and collectibles). Predicate triggers include **time predicates**: an object becoming older than a given threshold is a clock-fired applicability event. The layers of the architecture are:

Layer	Function

L0 Identity	One unique product identifier per subject (ISO/IEC 15459, EN 18219), with a granularity model of type, batch and item; bridges to GS1, Handle, Ecode and MA schemes; one physical carrier.
L1 Neutral core	A deliberately minimal passport skeleton: product and passport identifiers, status, timestamps, profile manifest, event log, signatures. No jurisdiction's fields live in the base. EN 18223's `contentSpecificationIds` is the EU's proto-manifest; the framework generalizes it.
L2 Profiles	The lens objects described above, composed on the three axes.
L3 Semantic registry	Every data point defined once in a federated register per ISO 19135:2025, seeded with UNTDED (paper 2).
L4 Trust	Per-jurisdiction trust frameworks, cross-recognized through trust lists and a multi-log master list; profile-bound cryptographic suites including SM, FIPS and post-quantum families (paper 5).
L5 Resolution	Resolver linksets (RFC 9264) keyed by profile context, role, language and request region — one carrier, many destinations.
L6 Regional serving	In-country DPP service provision (the EN 18221 pattern) where data-residency law requires it.

> **Neutrality is enforced at the mount, not asserted.** EN 18223 §0.3 declares the model "technology, ontology and domain agnostic" and EN 18216's scope requires operation "without vendor lock-in". The framework implements those stated goals: identity, semantics, trust and the profile interface are open ISO/IEC/UN instruments, and any proprietary metamodel — including the Asset Administration Shell — is confined to a single informative mapping (invariant I11).

Where jurisdictions regulate differently (for example divergent substance-restriction lists), they register **distinct data elements harmonized to a common concept**; disagreement between profiles is a correct outcome of projecting one canonical record under different rules, and is auditable — never silently merged (I2).

Capability classes and the passive→active twin axis

Most products in DPP scope (textiles, mattresses, most batteries) are silent objects: no connectivity, no keys. The framework fixes an explicit capability taxonomy and gates profile requirements by class: **S0 silent** (testimony-only); **S1 passive-auth** (NFC chip, PUF or IEC 61406 identity link — identity attestation without logs; the tier EN 18246 explicitly excluded from scope, a named gap); **S2 logged-contact** (dumps on physical read, no communications); **S3 connected** (full edge segments). Freshness and verification requirements must be satisfiable by the class — demanding live freshness from an S0

product is an unsatisfiable profile. The twin is accordingly **testimonial by default, sensorial by exception**, and a silent S0 product must be fully servable — through to the Tier-A minimum — with zero device participation ever.

The deployed national services are the predecessor of the DPP: the **passive-twin era**, where state exists only as point-in-time attestations and staleness between them is unbounded and undetectable. The **active twin** self-reports under device keys: freshness becomes bounded and detectable, and the binding question shifts from *who saw it* to *who attests the sensor*. Passive twin maps to S0–S1, active twin to S2–S3; both coexist permanently, with profiles declaring per data element the required **truth mode** — attest-sampled, self-committed, or both with precedence.

05 · PROPOSAL

What we propose JTC 5 adopt

The consortium proposes that JTC 5:

1. **Adopt the fourteen invariants as design constraints** for all JTC 5 deliverables, as the check-list against which base-model and profile proposals are reviewed.
2. **Establish a profile-mechanism standard as the flagship work item** — the registered profile object, the manifest schema, the mount, and the as-of applicability machinery — of which the semantic registry (paper 2) is one component. This is the deliverable that makes "your regulation becomes a governed profile, not an extension of another jurisdiction's envelope" operational, and gives every national body a symmetric seat at the table.
3. **Support the Part 1 / Part 2 split of AWI 25534-1** and accept the harvested terminology dataset (paper 6) as input to Part 1, resolving the definitional conflict of §2.1 before Part 2 work proceeds on contested ground.
4. **Fill the EN 18223 §4.3 vacancy** by taking the semantic registry requirements (paper 2) into the relevant workstream as an NWIP, with ISO/TC 154, UN/CEFACT and the FERIN operators as co-sponsors.
5. **Adopt the lifecycle event taxonomy and state machine** (paper 3) as the normative answer to the EN 18223 §4.2 mandate, on the ISO 8601 time basis already normatively referenced by the series.
6. **Take the SIGNATIF cross-recognition architecture** (paper 5) as the workstream D trust input, matching the governance sketch already present in AWI 25534-1, with CNML as the first demonstrated sector realization.
7. **Register implementation experience** (paper 4) as standing input: conformance methodology, corpus-derived negative test cases, and open reference code.

Sequencing proposed to the plenary: terminology and the registry first (they unblock everything else and are prerequisites of the profile mechanism); lifecycle and trust in parallel; the twin-projection doctrine and vertical pilots (OIML measuring instruments first) following in 2027, ahead of the EU registry implementing act (30/06/2027) and steel delegated act (30/07/2027).

06 · REFERENCES

Normative and informative citations

1. EN 18216:2026; EN 18219; EN 18220; EN 18221; EN 18222; EN 18223; EN 18239; EN 18246 — Digital Product Passport series, CEN-CLC/JTC 24.
2. AWI ISO 25534-1, *Digital product passport* — Part 1 (vocabulary) / Part 2, ISO/IEC JTC 5.
3. ISO 8601-1 and ISO 8601-2 (including Amd 1), date and time representation.
4. ISO/IEC 15459 series, unique identifiers; ISO 19135:2025, geographic information — procedures for item registration (FERIN).
5. UNTDED, United Nations Trade Data Element Directory (ISO/TC 154 custody).
6. RFC 9264, Linkset; ISO 10303-11 (EXPRESS); ISO 10303-242 application protocol.
7. ICAO Doc 9303, Machine Readable Travel Documents (CSCA master-list precedent).
8. ISO 20022; DCAT-AP; ETSI TS 119 612 (trusted lists).
9. Companion papers 2–6 of this series, UniDPP contribution set, 2026-09.

Joint drafting consortium: CalConnect (convenor), ISO/TC 154, OIML, ELF — in the UniDPP framework (multi-body stewardship; Apache-2.0/MIT per repository).

Paper 1 of 6 · an-international-framework-for-the-dpp.html · 2026-09-07